

15	A	The Barton's pendulums experiment demonstrates the physical phenomenon of resonance and the response of pendulums to oscillation at, below and above their resonant frequencies. This system vibrates at the resonance frequency of a driver pendulum, causing the target pendulum(s) to swing with the maximum amplitude. When comparing polystyrene and copper, the most striking difference is their density. Copper is approximately eight and a half times denser than polystyrene. This means that for pendulum bobs of the same size, the copper bob will be much heavier than the polystyrene one. This density difference has profound effects on how each pendulum behaves during the experiment. The polystyrene pendulum, being much lighter, is significantly affected by air resistance. As it swings through the air, the drag forces act more strongly relative to its small mass, causing it to lose energy quickly. This makes the polystyrene pendulum more heavily damped. In contrast, the copper pendulum has much greater mass packed into the same volume. When it swings, its high momentum means that air resistance has relatively little effect on its motion. The copper pendulum can maintain its oscillations for much longer periods and is less affected by external disturbances. When pendulum Z drives both pendulums through forced oscillations, these density differences become very apparent in the resonance behaviour. Hence, at resonance, the amplitude of oscillation of copper pendulum X will be greater than that of polystyrene pendulum Y.
16	B	At compression region of a wave travelling towards the right, the adjacent